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| **Fiscal Year:** | FY 2022 | **Task Last Updated:** | FY 05/05/2022 |
| **PI Name:** | Jaing, Crystal Ph.D. | | |
| **Project Title:** | International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) Project | | |
| **Division Name:** | Human Research, Space Biology | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | SPACE BIOLOGY--Cellular and molecular biology | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | (1) Cell & Molecular Biology<br>(2) Microbiology | | |
| **Space Biology Cross-Element Discipline:** | (1) Immunology | | |
| **Space Biology Special Category:** | (1) Translational (Countermeasure) Potential | | |
| **PI Email:** | jaing2@llnl.gov | **Fax:** | FY |
| **PI Organization Type:** | GOVERNMENT | **Phone:** | 925-424-6574 |
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| **Zip Code:** | 94550-9698 | **Congressional District:** | 15 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2014 Space Biology Flight (ILSRA) NNH14ZTT002N |
| **Start Date:** | 05/01/2015 | **End Date:** | 12/31/2021 |
| **No. of Post Docs:** | 3 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 0 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Loftus, David | **Contact Phone:** | 650-604-1011 |
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| **Flight Program:** | ISS | | |
| **Flight Assignment:** | NOTE: End date changed to 12/31/2021 per F. Hernandez/ARC (Ed., 11/19/20)<br>NOTE: End date changed to 12/31/2020 per PI and NSSC information (Ed., 11/26/19)<br>NOTE: End date changed to 12/31/2019 per F. Hernandez/ARC (Ed., 4/18/19)<br>NOTE change in grant number and end date per NSSC information and F. Hernandez/ARC; note grant 80NSSC18K0113 has period of performance of 10/20/2017-4/30/2019 (Ed., 1/18/2018)<br>NOTE: Extended to 3/31/2019 per F. Hernandez/ARC (Ed., 8/22/17) | | |
| **Key Personnel Changes/Previous PI:** | No changes. | | |
| **COI Name (Institution):** | Mehta, Satish Ph.D. ( KBR/NASA Johnson Space Center )<br>Pierson, Duane Ph.D. ( NASA Johnson Space Center )<br>Smith, David Ph.D. ( NASA Ames Research Center )<br>Venkateswaran, Kasthuri Ph.D. ( NASA Jet Propulsion Laboratory ) | | |
| **Grant/Contract No.:** | 80NSSC18K0113 ; NNX15AJ29G | | |
| **Performance Goal No.:** | | | |

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| **Performance Goal Text:** | |
| | [Ed. note (January 2018): change in grant number and end date per NSSC information and NASA Ames Research Center; note grant 80NSSC18K0113 has period of performance of 10/20/2017-4/30/2019; Ed. update 12/11/2020--end date is now 12/31/2021]<br>International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with various ISS modules using traditional culture-based methods and state-of-the art molecular techniques. The objectives of the ISS-MOP project are to measure presence of viral and select bacterial and fungal pathogens and correlate their presence on crew. The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration.<br><br>The hypothesis of the ISS-MOP project is that certain short-living viruses associated with inert environmental spaces in the closed habitat of the ISS can have an adverse influence on crew health. To test this hypothesis the team will catalog viruses associated with surfaces and air of the ISS modules using the all-encompassing Lawrence Livermore Microbial Detection Array (LLMDA), which contains probes to detect not only viruses but also clinically important human bacterial and fungal pathogens. In addition, skin and saliva samples from crewmembers will be requested to test the hypothesis. |
| **Task Description:** | There are three aims proposed in this ISS-MOP project. While the cultivation-based assay (Aim 1) provides a valid estimation of microbial presence, it offers only a limited assessment of the phylogenetic/pathogenic diversity and physiological breadth of the microbial population present. Therefore, the team will use the novel and comprehensive LLMDA microarray, which is specifically designed to elucidate the pathogenic viral and microbial diversity profiles of complex samples to at least the species level (Aim 2). In addition, we will characterize the virulence and antibiotic resistance mechanisms associated with these pathogens from the ISS by developing a new Virulence and Antibiotic Resistance gene array and analyze the samples on this array. The final deliverable of the proposed project to NASA will be the ISS-MOP dataset, a compilation of all of the genomic sequences and genetic information of viruses and microbes encountered on and within the ISS habitat (Aim 3).<br><br>The objectives of the ISS-MOP project directly respond to the National Research Council Committee for the Decadal Survey on Biological and Physical Sciences in Space recommendation P1 where it states, “to establish a microbial observatory program on the ISS to conduct long-term, multigenerational studies of microbial population dynamics.” The ISS-MOP project will also provide answers to NASA Space Biology Guiding Questions: How do alterations in gravity affect in microbial cells (CMM1); Does the enhanced virulence or other genetic changes observed with microbial cultures flown in space occur in other species (CMM4); and What are the genetic changes underlying physiological responses to spaceflight demonstrated in biological systems (CMM5). |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The effects of the spaceflight environment on viral and microbial pathogen dynamics are largely unknown and represent both a significant gap in our knowledge and an important opportunity to study the presence of microbial populations and predict health risks during long-term space exploration. International Space Station – Microbial Observatory of Pathogenic Virus, Bacteria, and Fungi (ISS-MOP) project will generate the microbial censuses associated with crewmembers and various ISS modules using both traditional culture-based methods and state-of-the art molecular techniques to a degree that has not previously been possible. |
| **Task Progress:** | The International Space Station (ISS) is a uniquely enclosed environment that has been continuously occupied for the last two decades. Throughout its operation, protecting the health of the astronauts on board has been a high priority. The human microbiome plays a significant role in maintaining human health, and disruptions in the microbiome have been linked to various diseases. To evaluate the effects of spaceflight on the human microbiome, body swabs and saliva samples were collected from four ISS astronauts on consecutive expeditions. Astronaut samples were analyzed using shotgun metagenomic sequencing and microarrays to characterize the microbial biodiversity before, during, and after the astronauts’ time on board the ISS. Samples were evaluated at an individual and population level to identify changes in microbial diversity and abundance. No significant changes in the number or relative abundance of taxa were observed between collection time points when samples from all four astronauts were analyzed together. When the astronauts’ saliva samples were analyzed individually, the saliva samples of some astronauts showed significant changes in the relative abundance of taxa during and after spaceflight. The relative abundance of Prevotella in saliva samples increased during two astronauts’ time on board the ISS, while the relative abundance of other commensal taxa, such as Neisseria, Rothia and Haemophilus, decreased. The abundance of some antimicrobial resistance genes within the saliva samples also showed significant changes. Most notably, elfamycin resistance gene significantly increased in all four astronauts post-flight and a CfxA6 beta-lactam marker significantly increased during spaceflight but returned to normal levels post-flight. The combination of both shotgun metagenomic sequencing and microarrays showed the benefit of both technologies in monitoring microbes on board the ISS. There were some changes in each astronaut’s microbiome during spaceflight, but these changes were not universal for all four astronauts. Two antimicrobial resistance gene markers did show a significant change in abundance in the saliva samples of all four astronauts across their collection times. These results provide insight for future ISS microbial monitoring studies and targets for antimicrobial resistance screenings. The microorganisms present on ISS surfaces originate from crew and cargo or from life support recirculation in an almost entirely closed system. The Microbial Tracking 1 (MT-1) project was the first ISS environmental surface study to report on the metagenome profiles without using whole genome amplification. The study surveyed the microbial communities from eight surfaces over a 14-month period. The Microbial Tracking 2 (MT-2) project aimed to continue the work of MT-1, sampling an additional four flights, over another year and a half. Eight surfaces across the ISS were sampled with sterile wipes and processed upon return to Earth. DNA, extracted from the processed samples (and controls) were treated with propidium monoazide to detect intact/viable cells, or left untreated, to detect the total DNA population (free DNA/compromised cells/intact cells/viable cells. Treated and untreated DNA were analyzed; metagenomics analysis was also conducted on the DNA samples. Culture and quantitative polymerase chain reaction (qPCR) analyses were also performed to supplement the metagenomics results. The overall number of microbial species detected at each time point did not significantly change, but species profiles did. Multi-dimensional scaling and |

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| | differential abundance analysis showed significant changes in the microbial populations between sampling time points but not across surface locations. The observed antimicrobial resistance gene profiles for the ISS surface samples were more stable over time, both temporally and spatially, with 29 antimicrobial resistance genes detected across all samples. Metagenomic assembled genomes were able to be reconstructed from the dataset, resulting in 82 metagenomic assembled genomes, which provided further insight into the functional capabilities of the microbiome of this unique built environment. The populations of bacteria and fungi collected from the eight ISS surfaces remained similar between locations but changed over the course of a year and a half. The microbial communities were dominated by organisms associated with the human skin, indicating a large exchange between the astronaut crewmembers and the ISS surfaces. Despite the changes in the microbial community, the presence of antimicrobial resistance genes was stable, indicating the persistence of genetic redundancy. |
| **Bibliography Type:** | Description: (Last Updated: 04/07/2023) |
| **Articles in Peer-reviewed Journals** | Morrison MD, Thissen JB, Karouia F, Mehta S, Urbaniak C, Venkateswaran K, Smith DJ, Jaing, C. "Investigation of spaceflight induced changes to astronaut microbiomes." Front. Microbiol. 2021 June 2;12:1368. https://doi.org/10.3389/fmicb.2021.659179 ; PMID: 34149649; PMCID: PMC8207296 , Jun-2021 |
| **Articles in Peer-reviewed Journals** | Simpson AC, Urbaniak C, Bateh JR, Singh NK, Wood JM, Debieu M, O'Hara NB, Houbraken J, Mason CE, Venkateswaran K. "Draft genome sequences of fungi isolated from the International Space Station during the Microbial Tracking-2 experiment." Microbiol Resour Announc. 2021 Sep 16;10(37):e0075121. https://doi.org/10.1128/MRA.00751-21 ; PMID: 34528817; PMCID: PMC8444978 , Sep-2021 |
| **Articles in Peer-reviewed Journals** | Simpson AC, Urbaniak C, Singh NK, Wood JM, Debieu M, O'Hara NB, Mason CE, Venkateswaran K. "Draft genome sequences of various bacterial phyla isolated from the International Space Station." Microbiol Resour Announc. 2021 Apr 29;10(17):e00214-21. https://doi.org/10.1128/MRA.00214-21 ; PMID: 33927037; PMCID: PMC8086211 , Apr-2021 |
| **Articles in Peer-reviewed Journals** | Urbaniak C, Morrison MD, Thissen JB, Karouia F, Smith DJ, Mehta S, Jaing C, Venkateswaran K. "Microbial tracking-2, a metagenomics analysis of bacteria and fungi onboard the International Space Station." Microbiome. 2022 Jun 29;10(1):100. https://doi.org/10.1186/s40168-022-01293-0 ; PMID: 35765106; PMCID: PMC9241228 , Jun-2022 |